Unit IV: MOBILE NETWORK LAYER

Mobile IP allows a device to **move between networks while keeping the same IP address**

Mobile IP Requirements

Requirement	Purpose	Example
Transparency	Apps don't notice mobility	Video streaming continues
IP Compatibility	Works with existing IP networks	No router changes
Location Transparency	Keeps same home IP	Mobile node moves to new network
Scalability	Efficient for many nodes	Large corporate network
Security	Protect registration & routing	Authenticate updates
Robustness	Seamless mobility	Minimize packet loss
Handoff Support	Smooth transition between networks	Wi-Fi to LTE handoff

Term	Meaning	Example
Mobile Node (MN)	Moving device	
Home Network	Original network	TN BSNL
Home Agent (HA)	Tracks MN location	
Foreign Network	Visiting network	Karnataka BSNL
Foreign Agent (FA)	Helps MN in foreign network	

Care-of Address (CoA)	Temporary address in new network
Correspondent Node (CN)	Communication partner

3. IP Packet Delivery

Without Mobile IP:

- Packets go to **home network → lost With Mobile**

IP:

1. Packet goes to **Home Agent**
2. HA forwards to **Care-of Address**
3. Delivered to mobile node This creates **triangle routing**:

CN → HA → MN

4. Agent Discovery

How MN knows it moved:

- **Agent Advertisement**
  - Broadcast by HA/FA (Towers continuously announcing Im the BSNL agent here, so mobile with auto signal detection attracted/registered ● **Agent Solicitation**
  - MN requests info if none received (with Manual tower detection, we search and connect to listed/interested service provider)

MN determines:

- Am I home or foreign? ● What is my CoA?

5. Registration MN must inform HA:

Steps:

1. MN gets CoA
2. Sends registration request (via FA)
3. HA updates location
4. HA sends reply

Includes authentication → prevents attacks

CoA types:

- Foreign Agent CoA: (tmp FA IP shared by few other guest devices in that new loc)

- Co-located CoA: (tmp unique IP for every guest device)

6. Tunneling & Encapsulation Used to forward packets.

Concept:

- Original packet is **wrapped inside another packet**

Types:

- IP-in-IP
- Minimal encapsulation
- GRE (Generic Routing Encapsulation) Structure:

Outer Header (HA → CoA)
Inner Header (Original CN → MN)

1. CN sends packet → MN (Home Address)
2. Packet reaches Home Agent (HA)
3. HA encapsulates the packet
4. HA sends it to CoA
5. Packet is decapsulated
6. MN receives original packet

7. **Optimizations**

Problem: Triangle routing inefficient

Solution:

- **Route Optimization**
  - CN sends directly to MN's CoA Requires:
- Binding cache (CN stores MN location)

8. **Reverse Tunneling** is a mechanism in Mobile IP where packets from a mobile node in a foreign network are encapsulated and sent to the Home Agent before being forwarded to the destination. It is used to overcome issues like ingress filtering and firewall restrictions by ensuring that packets appear to originate from the home network.

Used when:

- Firewalls block foreign addresses **Flow:**

MN → HA → CN
Ensures packets appear from home
network **9. Mobile IP in IPv6** IPv6
improves mobility:

- No need for Foreign Agent
- Built-in **auto-configuration**
- Route optimization is standard Key features:
- Binding Update
- Care-of Address auto-config

IPv4 Mobile IP had:

- Triangle routing problem
- Need for Foreign Agent (FA)
- Extra complexity MIPv6 improves by:
- Eliminating FA
- Supporting **direct routing**
- Using **auto-configuration**

1. MN moves to new network
2. Gets CoA (auto-config)
3. Sends Binding Update to HA
4. HA updates mapping
5. MN also informs CN (important!)
6. CN sends packets directly to MN

10. IP Micro-Mobility Support

Handles **movement within a local domain,** In normal Mobile IP (macro-mobility):

- Each movement requires updating Home Agent (HA) → causes delay and signaling overhead ● For frequent small movements, this is inefficient

Micro-mobility solutions handle local movements internally, only updating HA when moving out of the domain.
Goal:

- Reduce signaling to HA **Examples:**

1 Hierarchical Mobile IP (HMIP)

- Introduces **Mobility Anchor Point (MAP)**
- MN updates MAP instead of HA for local moves
- Reduces signaling and handoff latency

```
HA (Tamil Nadu)
   |
   ▼
Local MAP (Bengaluru)
   |
  ┌──────┬──────┐
  AR1   AR2   AR3
  MN moves: AR1 → AR2 → AR3
```

2 Cellular IP

- Designed for **wireless LANs and cellular networks**
- Uses **paging and routing tables** in access routers
- Handoff is **fast and localized**

3 HAWAII (Handoff-Aware Wireless Access Internet Infrastructure)

- Each domain has **foreign agent hierarchy**
- Optimized for multi-hop routing

Faster handoffs inside networks

11. Dynamic Host Configuration Protocol (DHCP)

Dynamic Host Configuration Protocol assigns IP addresses dynamically.

Process (DORA):

1. **Discover** – MN broadcasts a request for IP configuration
2. **Offer** – DHCP server responds with an available IP and configuration
3. **Request** – MN selects an offer and requests the assigned IP
4. **Acknowledge** – DHCP server confirms the assignment Used for:

- Assigning Care-of Address

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12. Mobile Ad-hoc Networks (MANETs)
No fixed infrastructure.
Characteristics:

- Self-organizing
- Dynamic topology
- Multi-hop communication **13. Routing in MANETs** Major challenge:
- Nodes move → routes break frequently

A. DSDV (Destination-Sequenced Distance Vector)

- Based on Bellman-Ford
- Each node maintains **routes to all nodes**
- Updates periodically
- Uses **sequence numbers** to avoid loops

Sample fields in routing table:

Field	Description
Destination	Target node address
Next Hop	Neighbor node to forward packet
Metric (Hop Count)	Number of hops to reach destination
Sequence Number	Ensures freshness of route

Install Time / Update Flag Optional: for deciding old vs new info

Features:

- Periodic updates
- Always maintains routes Pros:

- Simple Cons:
- High overhead

B. DSR (Dynamic Source Routing), Ad-hoc On-demand Distance Vector (AODV)

- **On-demand routing**

Steps:

1. Route Discovery
2. Route Maintenance Entire route stored in packet header Pros:
   - No periodic updates Cons:
   - Large headers

Sample routing table of Node A

Destination	**Route**	**Hop Count**
B	A → B	1
C	A → B → C	2
D	A → B → C → D	3

Feature	**DSDV**	**DSR**
Type	Proactive	Reactive
Route info	Maintained in table	Carried in packet headers
Overhead	High (periodic updates)	Low (on-demand)
Delay	Low (route ready)	Higher (route discovery first)

14. Alternative Metrics

Instead of shortest path:

- Energy consumption
- Link quality
- Delay
- Stability

Helps improve performance in mobile environments

Metric	Goal	Advantages	Disadvantages	Typical Use Case
Shortest Path	Minimize hop count	Simple, fast computation	May ignore link quality, energy, or delay	Traditional wired networks, basic routing protocols
Energy Consumption	Minimize energy used	Extends network lifetime, balances load	May result in longer paths or higher latency as this avoids routes with battery low hops	Wireless Sensor Networks (WSNs), IoT
Link Quality	Maximize reliability, Prefer links with higher reliability and lower error rates, even if the path is longer. **Metrics often used:** Signal-to-noise ratio (SNR), packet delivery ratio (PDR), or bit error rate (BER)	Reduces packet loss, retransmissions	May not be shortest or fastest	Wireless Mesh Networks, MANETs

Delay	Minimize end-to-end latency	Supports real-time applications, if shortest path takes only 2 hops with 100 ms as congested and other route has 4 hops with 80 ms, second route is opted	May consume more energy or bandwidth	VoIP, video streaming, control networks
Stability	Maximize path lifetime	Fewer route breaks, less overhead	Might not be optimal in distance or delay	VANETs, mobile ad hoc networks

15. Overview of Ad-hoc Routing Protocols

Type	Protocol	Behavior
Proactive	DSDV	Always maintains routes
Reactive	DSR, AODV	Finds routes when needed
Hybrid	ZRP	Combination

Zone Routing (ZRP)rotocol

Network Zones

- Each node defines a **zone** around itself with radius r hops.
- Inside this zone, the node knows **routes to all other nodes proactively** (like a table).

Proactive (Intra-zone)

- Uses **Intra-zone Routing Protocol (IARP)** to maintain up-to-date routes within the zone.
- Immediate route information is available for nearby nodes.

Reactive (Inter-zone)

- For nodes **outside the zone**, ZRP uses **Inter-zone Routing Protocol (IERP)** to discover routes **on-demand**.
- Reduces unnecessary flooding in the network.